COUNTERFACTUAL ATTRIBUTION AND BUDGET-AWARE DIAGNOSIS OF CASCADED INFORMATION RETRIEVAL

Revised Research Proposal and Methodology Guide

CS4.406 - Information Retrieval & Extraction
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Core idea: measure stage effects, test budget-controlled candidate allocation, and predict those effects from inexpensive signals and compact representations.

Executive Summary

Modern retrieval systems use a cascade: a fast first-stage retriever selects a limited candidate set, and a more expensive second-stage model reranks those candidates. Stage 1 controls eligibility; Stage 2 controls ordering among eligible documents. The project does not decompose a final score into additive stagewise score contributions. Instead, it measures how document-level ranking utility changes under explicitly defined pipeline interventions.

The revised proposal combines four elements. First, a counterfactual reference framework compares the actual cascade with a no-reranker world and, on a feasible reference subset, a no-gate world over a frozen evaluation universe. Second, a budget parameter t allocates a fixed reranker budget between documents retained from the Stage-1 candidate set and documents selected from the discarded set. Third, conventional ranking signals are the primary cheap-prediction baseline; LSH/neural-hashing features are an exploratory extension only if the exact method is confirmed and feature extraction improves the cost-fidelity frontier. Fourth, experiments vary stage mismatch, beginning with lexical BM25 versus dense semantic retrieval while holding the semantic reranker fixed.

Important distinction The full-universe no-gate effect and the t-based intervention are different estimands. The former removes the candidate restriction within a fixed universe; the latter reallocates a fixed reranker budget between candidates and outsiders. The t experiment is the practical budget-aware study, not a mathematical substitute for the full no-gate counterfactual.

The intended output is a reproducible benchmark and diagnostic methodology that measures stage effects, investigates when candidate allocation matters, and tests whether inexpensive signals can predict these effects accurately enough to save computation.

Document map

- Research question and contribution
- Cascade definitions and document-level utility
- Budget-controlled t intervention
- Stage mismatch axes
- Cheap signals and LSH/neural hashing
- Experiments, datasets and metrics
- Design freeze, risks and compute plan
- Three-member parallel work split
- Shared interface contracts and pre-run checklist

1. Research Problem and Questions

A poor final result in a cascaded retrieval system can arise because a useful document never reaches Stage 2 or because Stage 2 gives a poor position to a document it did receive. These failure modes call for different interventions: improving recall or candidate depth versus improving the reranker. Aggregate NDCG alone does not localize the cause, and raw rank disagreement is not a stagewise causal contribution.

Primary question Can we quantify document-level effects of candidate gating and reranking, measure how these effects change when a fixed reranking budget is shared with discarded documents, and predict the effects from inexpensive signals?

1.1 Research questions

1. RQ1 - Counterfactual attribution: How does document utility change when the reranker is removed or the candidate gate is removed within a fixed universe?

2. RQ2 - Budgeted candidate allocation: Under a fixed reranker budget, how does the utility change as a larger fraction of reranker inputs is drawn from documents discarded by Stage 1?
3. RQ3 - Cheap prediction: Can conventional ranking features predict the reference effects and t-allocation outcomes, and do compact hash-derived features add enough predictive value to justify their extraction cost?
4. RQ4 - Mismatch: Do effects and prediction accuracy change when Stage 1 and Stage 2 differ in retrieval signal, ranking objective or computational cost?

1.2 What the project is not

- It is not a score-fusion model such as final_score = alpha*s1 + (1-alpha)*s2. That is a different architecture.
- It does not claim that Stage-1 rank or score flows into the Stage-2 score. In the core pointwise setup, Stage 2 scores each query-document pair independently.
- It does not claim to recover full-corpus outcomes from a limited evaluation universe. All no-gate results are explicitly universe-relative.
- It does not assume hashing features are cheap or useful in advance; feature extraction cost and prediction value are measured.

2. Cascade Model and Reference Worlds

For a query q, let U_q be a frozen evaluation universe assembled from specified retrieval runs. A Stage-1 system computes score s1(q,d) for d in U_q and selects C_q = Top-Kc(U_q,s1). The discarded set is D_q = U_q minus C_q. Stage 2 uses a pointwise scoring function s2(q,d)=f2(q,d) and sorts the admitted candidates. No Stage-1 score, rank, source label or input ordering is fed into the Stage-2 model.

World	Definition	What it isolates
A. Actual cascade	Stage 1 selects C_q; Stage 2 reranks C_q.	Observed baseline system.
B. No reranker	Keep exactly C_q, but return Stage-1 ordering within C_q.	Effect of replacing Stage 2 while preserving the gate.
C. No gate within U_q	Stage 2 ranks every document in the frozen U_q.	Effect of removing the gate within this fixed universe.
D. t-budget world	Rerank B documents, with tB from C_q and (1-t)B from D_q.	Budget-constrained candidate reallocation; distinct from World C.

The universe is a fixed, reproducible reference set rather than the entire corpus. A candidate set must be a subset of U_q for every core Stage-1 model. If U_q is identical to C_q, the no-gate effect is zero by construction, so the universe must be built independently of the candidate cutoff.

2.1 Proposed universe construction

A provisional construction is the deduplicated union of top-M results from BM25, DPR and SPLADE-v3. The original proposal uses M=5,000 per source. BM25 and DPR form the main Stage-1 comparison; SPLADE-v3 is optional if indexing and compute permit. Every Stage-1 retriever used in a core comparison must contribute its top-M results to U_q before candidates are selected. If another retriever is added, regenerate U_q.

2.2 Pointwise reranker assumption

For the core design, s2(q,d) is a function of the query-document pair only. For a survivor d, its Stage-2 score is unchanged when the set expands from C_q to U_q; only the competing documents change. This gives the gate intervention a clean competitive-set interpretation. A listwise reranker can depend on the complete candidate set and should be a separate extension because its gate effect also includes contextual reranker changes.

3. Document Utility and Counterfactual Effects

Use the same relevance-based document utility in all worlds. The primary choice is a per-document DCG@k contribution: U_d(pi)=G(rel_d)/log2(1+rank_pi(d)) when rank is within k, and zero otherwise. Gains must follow the dataset-specific qrel semantics. In TREC Deep Learning passage ranking, grade 1 means “Related” rather than relevant; use zero gain for grades 0 and 1, and a prespecified graded mapping for grades 2 and 3 (for example, gains 3 and 7). Do not blindly apply G(rel)=2^rel-1 to grade 1. For other qrels, document and freeze the gain mapping. Top-k inclusion and RBP-style utility can be used for sensitivity analysis.

Effect	Definition	Interpretation
Reranking effect	E_rerank(q,d) = U_d(pi_actual) - U_d(pi_no-reranker)	How Stage 2 changed the document outcome given the same C_q.
Gate effect	E_gate(q,d) = U_d(pi_actual) - U_d(pi_no-gate)	How the gate changed the outcome relative to Stage 2 ranking all of U_q.
Budget-allocation effect	Delta_t(q,d) = U_d(pi_t) - U_d(pi_t=1)	How a t-controlled allocation changes utility relative to the candidate-only allocation at the same budget.

These are signed effects and do not form an additive decomposition. Under a pointwise reranker and monotone rank utility, E_gate is non-negative for survivors because expanding the competition can only leave their rank unchanged or push them down; for non-survivors it is non-positive because their actual utility is zero and the no-gate world may give them positive utility. Therefore, all gate-effect prediction results must be stratified by survivorship, and predictors should be evaluated on effect magnitude within each stratum to avoid trivially learning candidate membership.

Reference targets The effects are operationally defined by the declared intervention and metric. On real data they are reference effects conditional on the frozen universe and available qrels, not universal causal truth about the full corpus.

4. Supervisor Proposal: Fixed-Budget t Allocation

The full no-gate world requires Stage 2 scores for every document in U_q and may be expensive. The new practical experiment controls the reranker budget and varies the source composition of its inputs. Let B be the total number of documents sent to the reranker in one query. For each t in a pre-registered grid, choose floor(tB) documents from the Stage-1 candidate set C_q and B-floor(tB) documents from the discarded set D_q, then use the same Stage-2 model to rank their union.

Design constraint For t=1 to be the conventional cascade baseline, set B=Kc for that experiment and include all C_q. If B differs from Kc, t=1 is only a candidate-only budget baseline, not the full actual cascade. State this distinction whenever results are reported.

4.1 Allocation grid and outsider selection

t value	Reranker inputs from C_q	Reranker inputs from D_q	Interpretation
1.00	B	0	Candidate-only baseline
0.90	0.90B	0.10B	Small outsider allocation
0.75	0.75B	0.25B	Moderate outsider allocation

t value	Reranker inputs from C_q	Reranker inputs from D_q	Interpretation
0.50	0.50B	0.50B	Equal allocation
0.25 / 0.00 (optional)	25% / 0%	75% / 100%	Stress-test candidate replacement

Recommended outsider selection policies should be evaluated separately: (i) boundary-local outsiders immediately below the Stage-1 cutoff; (ii) documents retrieved highly by an alternative retrieval source; and (iii) random outsiders. The candidate portion should follow a fixed rule, such as the top tB documents by Stage-1 rank. This rank is used only to choose the submitted subset; it is not passed as a feature to Stage 2, which scores each query-document pair independently. Use multiple seeds for randomized policies and report their variation.

4.2 Outcomes and interpretation

- Compute NDCG@k or the chosen ranking metric for each t, plus per-document Delta_t.
- Measure relevant-document recovery from D_q and whether outsiders enter the final top-k. On queries with reference-world runs, define gate-recoverable gain G_q = DCG@k(pi_no-gate) - DCG@k(pi_actual), and reranking sensitivity R_q = |DCG@k(pi_actual) - DCG@k(pi_no-reranker)|. With a prespecified threshold tau, label a query gate-sensitive if G_q > tau and G_q > R_q; label it reranking-sensitive if R_q > tau and R_q >= max(G_q, 0); otherwise label it mixed/stable. Then test whether lower t yields greater Delta DCG on gate-sensitive than reranking-sensitive queries. This uses reference effects to define the diagnosis and a distinct budget-allocation intervention as the outcome; report that the analysis is conditional on U_q and judged qrels.
- Report reranker calls, latency and feature-generation cost at each t.
- If utility improves as outsider fraction increases, the current candidate set may be omitting useful documents under this allocation policy. If it falls, outsiders may be weaker or may displace better candidates.
- Poor outsider results do not prove the retriever is universally good: results depend on the sampled outsider pool, qrels coverage, model quality and utility function.

4.3 Relationship to the original no-gate world

The no-gate world ranks all of U_q with Stage 2, while a t-world replaces part of C_q with a budget-matched subset of D_q. Thus E_gate and Delta_t answer different questions. The recommended design is to retain exact no-gate effects on a manageable reference subset of queries/documents where compute allows, and use the t experiment as the main budget-aware intervention. The t experiment does not need full-universe inference at each t.

4.4 Membership-defined strata for Delta_t

For each t<1, compare each document's utility with its t=1 candidate-only baseline. Three of the four groups have signs determined by allocation membership, so pooled correlation can look strong even when a predictor learns only group membership. Primary Delta_t prediction metrics must therefore be reported within the following four strata, with a strata-only baseline that predicts the training-set mean Delta_t for each stratum (and zero for the structurally-zero group). Report macro-averaged within-stratum errors; pooled metrics may be included only as secondary summaries.

Document group at this t	Utility at t=1	Utility at t<1	Delta_t sign / evaluation
Candidate retained: in the top tB of C_q	May be positive	May be positive; competitors changed	Either sign; primary magnitude-prediction stratum
Candidate removed: in C_q but below the selected top tB	May be positive	Zero; not sent to Stage 2	Delta_t <= 0; report within-stratum magnitude error
Selected outsider: chosen from D_q	Zero; not in candidate-only baseline	May be positive	Delta_t >= 0; report within-stratum magnitude error
Unselected outsider: in D_q but not chosen	Zero	Zero	Delta_t = 0 by construction; report counts, not correlation

Implementation consequence: the shared effect table must validate signs conditionally on world_id/effect_type. E_gate sign checks and Delta_t sign checks have different invariants and must never share one unconditional validator.

5. Stage Mismatch as an Experimental Factor

The supervisor's mismatch idea is split into three distinct axes: mismatch in relevance evidence, mismatch in the ranking objective, and mismatch in computational cost. These are not interchangeable and should not be changed simultaneously in a single comparison. Retrieval-conditioned reranking is an established adjacent line of work [24]; this proposal studies explicit pipeline interventions and whether inexpensive features predict the resulting document-level effects.

Axis	Example pairing	Meaning	Status
Lexical vs semantic evidence	BM25 -> MiniLM cross-encoder	Stage 1 is driven by token overlap; Stage 2 uses contextual query-document interaction.	Core
Dense semantic vs semantic reranking	DPR -> same MiniLM reranker	Both use learned semantic representations but may have different training objectives and candidate behavior.	Core comparison
Learned sparse vs semantic	SPLADE-v3 -> same MiniLM reranker	Tests learned sparse lexical expansion against dense retrieval.	Optional core extension
Content vs behaviour	Content retriever -> click/purchase/engagement ranker	Different evidence/objective; behavioural labels may have exposure and position bias.	Stretch goal only
Cheap vs expensive computation	Lightweight reranker -> larger cross-encoder	Primarily a cost/capacity contrast, not necessarily objective mismatch.	Secondary axis

For the core comparison, change Stage 1 while holding Stage 2, U_q, query set, qrels policy and evaluation cutoff fixed. To study reranker cost/capacity, hold Stage 1 fixed and compare rerankers. Behavioural ranking is not required for the main project because it needs a suitable interaction dataset and a defensible bias-aware evaluation protocol.

5.1 Mismatch hypotheses

- Lexical-to-semantic cascades may have useful semantic outsiders that lexical candidate generation failed to retrieve; whether they improve utility depends on how those outsiders are selected and on the candidate budget.
- A dense retriever followed by the same semantic reranker may have different candidate overlap, recall and gate-effect distributions from BM25 followed by that reranker.
- Mismatch may change the predictability of attribution, but this is empirical rather than guaranteed.
- A more expensive Stage 2 may improve ordering quality yet still be constrained by a weak candidate pool; its gain should be evaluated jointly with compute.

6. Predicting Effects with Cheap Signals and Hash Features

The benchmark computes reference effects first, then asks whether features available from the factual execution can predict them. Conventional features form the required baseline. Hash-derived features are exploratory and will be retained only if their full generation cost improves the cost-fidelity frontier.

6.1 Conventional feature baseline

- Stage-1 rank, retrieval score and margin/distance to Kc.
- Candidate membership and candidate depth, with gate effects analysed separately within membership strata.
- Stage-2 score, rank and margin to the final top-k boundary for documents that were scored in the factual run.
- Within-candidate rank displacement computed by scoring the same C_q with Stage 1 and Stage 2; this is a predictor, not a causal measure.
- Alternative-retriever rank, source agreement and query-level signals if available without target leakage.

6.2 LSH/neural-hashing feature experiment

Exploratory only: test compact binary representations or hash-derived similarity features from both stages after the conventional predictor baseline is complete. A small hash is not inherently cheap: the experiment is justified only if representations are already available or can be generated at materially lower cost than the reference intervention, and if they improve the cost-fidelity frontier.

Feature set	Inputs	Question
F1 Conventional	Ranks, scores, margins, displacement	How far can ordinary logs predict effects?
F2 Stage-1 hash	Retrieval-side compact codes and similarities	Exploratory: does retrieval-side representation add value at acceptable extraction cost?
F3 Stage-2 hash	Reranker-side compact codes/features where available	Exploratory only; representation extraction cost may require expensive inference.
F4 Combined	F1 + F2 + F3	Exploratory: retain only if it improves the cost-fidelity frontier over F1.

Critical compute caveat A representation is not “cheap” merely because its final hash is small. If producing Stage-2 features requires an expensive cross-encoder pass for every outsider, include that inference in the cost. Prefer cached/available representations or a separately measured lightweight feature extractor.

6.3 Prediction targets

Evaluate predictors for E_rerank and E_gate on reference queries where those effects are computed. Separately train/evaluate predictors for Delta_t and query-level sensitivity across the t curve. Do not mix these targets into one label: a model that predicts the budget-allocation outcome has not automatically predicted the full no-gate effect.

6.4 Hashing implementation options

- If the hashing extension is retained, LSH over embeddings already produced by a retriever/encoder is the simplest first baseline; record the distance metric and number of bits.
- Neural hashing may learn compact codes for a specific similarity or downstream objective; fit it on training queries only and freeze it before held-out evaluation.
- Only if the extension passes the cost-feasibility check, compare Stage-1 features, Stage-2 features and their combination. Report feature extraction time, memory and prediction latency, not just MAE/correlation. A full neural-hashing ablation is not part of the minimum viable project.
- Confirm whether the intended method is LSH, LEH, or another named hashing method before implementation. Until the exact method and a low-cost feature-generation path are confirmed, treat hashing as an optional extension rather than a core dependency.

7. Synthetic Benchmark

Synthetic data provides exact reference effects and tests whether the implementation behaves as intended before real qrels and incomplete judgments complicate analysis. Generate an explicit latent relevance state rel_d and independent Stage-1 and Stage-2 deviations. Both stage scores may correlate through their shared relevance signal, but Stage 2 must not receive Stage-1 scores or ranks.

Vary Stage-1 quality, Stage-2 quality, noise levels, candidate depth, relevance preference alignment, and the strength of high-quality outsiders. Calculate the reference effects from the actual generated worlds rather than planting gate-sensitive/reranker-sensitive labels. In addition to the full no-gate reference, run t allocations to test whether the utility curve behaves as expected.

Synthetic regime	Manipulation	Purpose
Strong stages	Both stages align strongly with relevance	Check stable/easy conditions.
Weak retriever, strong reranker	Stage 1 noisier; Stage 2 stronger	Test gating as bottleneck.
Strong retriever, weak reranker	Stage 1 strong; Stage 2 noisier	Test reranker harm.
Gate stress	Some relevant documents rank just below or far below Kc	Test boundary probes and missed distant outsiders.
Stage mismatch	Vary stage preference/feature alignment without score passing	Test whether predictors generalize across pairings.
Budget stress	Vary t, B and outsider-selection policy	Test candidate replacement under fixed cost.

8. Real-Data Benchmark and Judgments

Use TREC Deep Learning 2019/2020 on the MS MARCO passage collection as the primary benchmark: passage-length documents fit the intended pointwise cross-encoder setting and align naturally with the neural passage-retrieval ecosystem. The qrels remain incomplete relative to a multi-retriever top-M union, so this choice is conditional on a qrels-coverage audit. Robust04 is a secondary sensitivity dataset for a small topic subset, not the default primary dataset: although its pooled judgments can help examine judgment-coverage sensitivity, it is a long-document newswire collection and does not come with a corresponding prebuilt dense index in the standard Pyserini setup [27].

Candidate dataset	Advantages	Risks / check
TREC Deep Learning 2019/2020 + MS MARCO passages	Primary benchmark: passage-length inputs fit the intended pointwise cross-encoder setup and the neural passage-retrieval ecosystem; standard queries and qrels are available.	Judgments remain sparse relative to a multi-retriever top-M union. Verify qrels coverage in U/C/D, compatible dense checkpoints/indexes, and held-out query feasibility before freezing.
TREC Robust04	Secondary small-subset sensitivity check for whether gate conclusions change in a collection with pooled document-level judgments.	Older newswire documents are long; Pyserini supplies a sparse Lucene index but not a drop-in dense index. A semantic comparison requires a verified dense checkpoint/index path; cross-encoder needs fixed document segmentation and aggregation (e.g., max-passage; PARADE is a more involved alternative [26]) with passage-level compute counted. Domain and granularity differ, so this is not a pure qrels-sparsity ablation.

Candidate dataset	Advantages	Risks / check
MS MARCO train/dev	Useful for model fitting and hashing/predictor training where needed.	Training labels are not a substitute for a held-out evaluation set; separate training, tuning and test queries and avoid evaluation leakage.

Before locking the main dataset, check both judgment coverage and model-artifact feasibility. For each query and K, compute: (i) qrel coverage of U_q, C_q and D_q; (ii) judged non-survivor counts; (iii) candidate overlap/Jaccard and Recall@K against available qrels; and (iv) whether a compatible Stage-1 checkpoint/index and Stage-2 input representation are actually available. TREC DL/MS MARCO is primary if these checks pass. Use Robust04 only as a secondary small-subset sensitivity analysis if a dense retrieval path can be run and long documents are segmented and aggregated with a frozen method such as max-passage scoring; count passage-level cross-encoder calls, not merely document count. A Robust04-vs-TREC-DL contrast is not a pure test of qrels sparsity because corpus, query and document granularity also change.

8.1 Incomplete judgments policy

- An unjudged document is not automatically irrelevant. Preserve a judged/unjudged flag in all document-level records.
- Use official qrels conventions for standard effectiveness metrics and clearly state the policy for per-document utility. For TREC DL passage ranking, grade 1 (“Related”) is not relevant; map grades 0 and 1 to zero gain for the primary utility, and freeze the gains for grades 2 and 3. Do not silently treat grade 1 as positive gain.
- Report judged-document coverage separately for candidate and discarded pools; do not make strong gate-failure claims when outsider judgments are insufficient.
- Use sensitivity measures such as RBP or restricted judged-pair analysis where appropriate, and explicitly acknowledge that no-gate outcomes remain universe-relative.

9. Experiment Plan

ID	Experiment	Main design	Main result
E1	Synthetic implementation validation	Known relevance; independent stage errors; all worlds and t grid.	Correctness, effect sanity, predictor recoverability.
E2	Core cascade and reranking effect	BM25 -> fixed pointwise MiniLM; compare Stage-1 ordering and Stage-2 ordering on same candidates.	E_rerank distribution and baseline predictor performance.
E3	Exact gate reference subset	Score fixed U for a feasible subset of queries; compare actual vs no-gate.	E_gate reference effects; validates local probes/predictors.
E4	t-budget allocation	At fixed B=Kc, vary t in {1, .9, .75, .5}; repeat prespecified outsider policies/seeds. On reference queries, define G_q=DCG(no-gate)-DCG(actual) and R_q=|DCG(actual)-DCG(no-reranker)|; test whether gate-sensitive queries gain more from lower t than reranking-sensitive queries.	NDCG/utility, outsider recovery, Delta_t, cost curves and the diagnosis-by-t interaction.
E5	Stage mismatch: lexical vs dense	BM25 and DPR use same U, K values, qrels and Stage-2 model.	Differences in effects and predictor accuracy.

ID	Experiment	Main design	Main result
E6	Optional exploratory hashing cost-fidelity ablation	Only after confirming exact LSH/LEH/neural-hashing method and inexpensive representation source; compare conventional features with Stage-1, Stage-2 and combined hash features.	Keep only if predictive gain improves the measured cost-fidelity frontier; otherwise document as infeasible/not pursued.
E7	Cost/capacity axis	Hold Stage 1 fixed; compare lightweight versus more expensive reranker if feasible.	Quality-compute frontier; not labelled as objective mismatch.
E8	Optional learned-sparse comparison	SPLADE-v3 as third Stage-1 system with same Stage 2.	Generalization across retrieval paradigms.
E9	Optional content-behaviour setting	Only if suitable interaction data and evaluation protocol are available.	Objective mismatch diagnosis; outside MVP.
E10	Optional listwise ablation	Fixed Stage 1; pointwise MiniLM versus ListT5/listwise reranker.	Whether candidate-context dependence changes gate effects.

Core priority: E1-E5. E6 hashing is exploratory and is promoted only if the exact method is identified and its feature-generation cost is demonstrably low or improves the cost-fidelity frontier. E7 is a secondary cost/capacity axis if compute permits; E8-E10 are extensions, not dependencies. Change one major axis at a time: when changing Stage 1, hold Stage 2 fixed; when changing Stage 2, hold Stage 1 fixed.

10. Hypotheses and Evaluation

Hypothesis	Test
H1: rank displacement is informative for reranking effects but insufficient for gate effects.	Compare displacement baseline to E_rerank and E_gate references, with gate results stratified by survivor status.
H2: a modest outsider budget can recover some value missed by the gate.	Measure utility and relevant outsider recovery across t and selection policies; expect effects to be query- and model-dependent.
H3: stage mismatch changes attribution patterns.	Compare BM25 -> MiniLM versus DPR -> MiniLM under identical U, K, qrels and cutoff.
H4: hash features can add predictive information.	Ablate Stage-1 hashes, Stage-2 hashes and combined features against conventional features, including extraction cost.
H5: low-cost probes can approximate exact gate effects.	Compare boundary, alternative-retriever and random outsider probes to full-universe effects on reference queries.
H6: more expensive reranking changes a quality-compute trade-off, not necessarily the objective mismatch.	Compare quality, attribution outcomes and latency while holding Stage 1 fixed.
H7: Reference attribution predicts which queries benefit from outsider allocation.	On the exact-reference subset, classify queries using G_q and R_q fixed above; compare Delta DCG across t for gate-sensitive versus reranking-sensitive groups. Use the t outcomes only as outcomes, not to define groups.

10.1 Metrics

- Ranking quality: NDCG@10 (primary), with top-k inclusion and RBP-style sensitivity if feasible.

- Effect prediction: MAE, Pearson/Spearman correlation, signed bias, top-m recovery of high-effect documents and calibration where meaningful.
- Budget experiment: utility vs t, relevant outsider recovery, actual reranker calls, latency and repeat-seed variation.
- Optional hashing extension: feature extraction latency, code size/memory, similarity computation time, and downstream prediction gain over the conventional baseline; keep it only if it improves cost-fidelity.
- Coverage/robustness: qrel coverage in U/C/D, number of judged non-survivors, candidate-set Jaccard/Recall@K, effect stability over K and seeds.

10.2 Split and leakage rules

- Split train/validation/test by query, not by query-document row.
- Fit neural hashing, feature normalization and prediction models on training queries only.
- Never use qrels, counterfactual worlds, future ranks, or target-derived labels as predictor input unless an explicitly labelled oracle upper bound is reported.
- For Stage-2 features, score only where those scores are available in the factual pipeline or account for extra inference required to make them available.

11. Common Design Choices to Freeze Before Main Runs

Choice	Required decision
Dataset/split	Primary: TREC DL 2019/2020 on MS MARCO passages, conditional on qrels-coverage and model-artifact checks; Robust04 only as a small secondary sensitivity check if dense retrieval and long-document aggregation are feasible. Split by query.
Universe U	Source retrievers, top-M per source, deduplication, tie-breaking and max universe size.
Stage-1 systems	BM25 and DPR core; SPLADE-v3 optional. Exact checkpoints/index settings recorded.
Stage-2 model	Pointwise reranker checkpoint, tokenizer, max length, truncation, precision, batch size and score cache schema.
Kc, B and t	Candidate depth Kc; set B=Kc for t=1 to equal full candidate-only baseline; freeze t grid and outsider-selection policies.
Reference worlds	Definitions for actual, no-reranker, no-gate and t-world; exact query subset used for full no-gate scoring.
Utility/qrels	Freeze gain mapping, cutoff k, qrels policy and unjudged-document treatment. For TREC DL passages, grades 0 and 1 get zero gain because “Related” is not relevant; freeze gains for grades 2/3. Report qrel coverage and incomplete-judgment sensitivity.
Hash features	Optional extension only until method is confirmed; record algorithm (LSH/LEH/other), representation source, bit length, training split, seed and full extraction cost. Require improvement to cost-fidelity frontier.
Mismatch conditions	Which comparisons alter evidence type, objective, or model cost; only one factor changed at a time.
Predictor/evaluation	Separate targets E_gate, E_rerank and Delta_t. Query-level split; survivor/non-survivor strata for E_gate; four membership-defined strata for Delta_t; compare against stratum-only baselines; bootstrap uncertainty by query.

Choice	Required decision
Compute/reproducibility	Hardware, environment versions, cache policy, seeds, logging, run names and output schemas.

12. Compute, Caching and Exit Checks

Pointwise scores can be cached by (query_id, doc_id, model_version). On passage collections, count one inference per passage scored. For Robust04 documents, a cross-encoder cannot consume the entire document in one 512-token input; any secondary experiment must predefine segmentation and document-level aggregation and must count the multiplied passage-scoring cost. Once cached, reusing scores to reconstruct rankings for different candidate depths or t policies does not require rerunning the model for repeated documents. The full no-gate world requires Stage-2 scores for every document in U_q, so run it only after throughput and coverage checks, and use a reference subset if needed. The t experiment scores only its B allocated documents per setting; repeated outsider policies/seeds increase compute and must be logged.

12.1 Pre-run validation checks

- Qrels coverage audit for U, C and D on each candidate dataset; report judged non-survivors at each candidate depth.
- Candidate-set overlap/Jaccard and Recall@K for BM25 vs DPR ; if sets are nearly identical, revise the interpretation or select a more informative pairing before main runs.
- Measure real cross-encoder throughput at the chosen sequence length, batch size and precision.

Retriever-artifact gate: verify that a compatible dense checkpoint and index/build path exists for the selected collection; verify document lengths are compatible with the reranker or define segmentation/aggregation and measure its cost. Dataset choice is not final until this check passes.

- Confirm C is always a subset of U and that the same fixed universe is used across Stage-1 comparisons.
- Validate that t=1 corresponds to the complete candidate-only baseline when B=Kc, and that every t allocation totals exactly B.
- Run pointwise shuffle-invariance test: permuting candidate input order does not change per-document Stage-2 scores.
- Estimate cost for full no-gate reference, t grid, hash features and outsider-probe baselines before expanding the query set.

12.2 Software and hardware

Use sequential retrieval/index runs where possible, store only top-M result IDs/scores to form U_q, and use cached pointwise scores for repeated rankings. A single-GPU workstation/cloud instance may suffice for MiniLM and feature models, but the actual throughput must be measured on the available hardware. Storage planning must include corpus subsets, indexes, run files, cached Stage-2 scores, hash codes and logs. Freeze package versions and report hardware/precision/batching with results.

13. Three-Member Parallel Work Split

The work is divided into three vertical streams. Each member can begin using toy data, synthetic outputs or cached runs. Stable file contracts are agreed first, so no member waits for another member's full model/index/pipeline.

Member 1 - Retrieval, universe and t-allocation runner

- Own corpus/query/qrels input adapters, IDs, dataset manifests and retrieval-run ingestion. Validate TREC DL/MS MARCO passage artifacts for the primary setup; separately assess whether Robust04 has a feasible dense retrieval path before treating it as a semantic-mismatch dataset.
- Run BM25 and a compatible dense retriever (DPR: castorini/cosdpr-distil model) on the selected passage collection; optionally SPLADE-v3. Do not assume a dense index exists for Robust04: obtain/build a compatible index or keep that collection to a limited lexical/qrels sensitivity analysis.

- Implement C and D construction, candidate overlap and retrieval recall audits, and t allocation policies for boundary, alternative-retriever and random outsiders.
- Export stable candidate/universe/allocation files and an executable toy fixture. This work can proceed before reranker integration by using cached or mock scores.

Member 2 - Utility, counterfactual engine and synthetic benchmark

- Own utility calculation, the actual/no-reranker/no-gate worlds, and the t-world ranking/Delta_t evaluator.
- Build synthetic data with explicit latent relevance and independent Stage-1/Stage-2 errors; create automated invariants and edge-case tests.
- Implement effect records keyed by query/document/world/K/t/seed; test that no-reranker preserves C and no-gate uses fixed U.
- Can validate all worlds with toy scores and toy qrels while Member 1 prepares real retrieval runs.

Member 3 - Reranker, hashing features and prediction evaluation

- Own pointwise semantic reranker wrapper, score caching and throughput benchmark.
- Implement the conventional feature baseline first. Add Stage-1 hash, Stage-2 hash and combined feature pipelines only as a cost-tested extension after confirming the exact hashing method and representation source; report feature extraction runtime and memory.
- Train/evaluate predictors for E_rerank, E_gate on reference queries, and Delta_t on budget runs. Report MAE, correlation, top-m recovery and cost.
- Can develop against synthetic effect tables and mock features before retrieval or full counterfactual runs are available.

14. Shared Interfaces and No-Blocking Protocol

Artifact	Minimum schema
Retriever run	query_id, doc_id, system_id, score, rank, checkpoint/index version
Universe/candidate file	query_id, doc_id, source_systems, in_U, in_C, Kc, Stage-1 score/rank, judged flag
Reranker run	query_id, doc_id, model_version, score, rank, inference settings
World/effect table	query_id, doc_id, world_id/effect_type, Kc, B, t, policy, seed, rank, utility, E_gate, E_rerank, Delta_t, t_stratum; validate sign invariants conditionally on world_id/effect_type.
Feature table	query_id, doc_id, feature_version, named features, hash method/version, feature-generation time
Evaluation report	dataset/split/config ID, metric, estimate, uncertainty, qrel coverage, runtime and environment

- Use stable query/document identifiers. Join records by ID rather than row order.
- Keep targets/effects separate from feature tables to prevent accidental leakage.
- Each member maintains one GPU-free smoke test and one minimal fixture.
- Use versioned CSV/JSONL/Parquet or agreed TSV schemas; never depend on importing another member's unfinished code.
- Hold regular coordination check-ins and maintain one design-decision log. If a model/index is delayed, continue with synthetic data, cached runs or mocks and mark the result provisional.
- Any change to U, Kc, B, t, qrels policy, utility, model checkpoint or hash method must be documented and accompanied by a config version.

15. Threats to Validity and Mitigation

Risk	Why it matters	Mitigation
Incomplete qrels	Outsiders may be unjudged; apparent gate benefit may be distorted.	Coverage audit; separate judged/unjudged; limit claims and run sensitivity analysis.
Universe dependence	E_gate only describes removal of the gate within U.	Freeze U; report its sources/size; do not claim full-corpus effect.
t policy dependence	Different outsider policies can give different outcomes.	Compare prespecified policies; multiple seeds; report policy-specific results.
Changing both stages	Makes it unclear whether effects are due to Stage 1 or Stage 2.	Hold Stage 2 fixed when changing Stage 1, and vice versa.
Hash features are not cheap	Stage-2 encoding may require expensive inference.	Count feature-generation compute and measure cost-fidelity, not prediction latency alone.
Membership shortcut	Gate effect sign is linked to survivor status under pointwise assumptions.	Stratify by survivorship and evaluate within-stratum magnitude.
Training/evaluation leakage	Query-level overlap inflates prediction results.	Split by query; train hashing/normalization on training data only.
Stage mismatch confounding	Checkpoint-domain shift may masquerade as paradigm mismatch.	Freeze and report training domains; choose matched models or explicitly label the confound.
Reranker set size	Scoring a much larger U may be outside intended deployment regime.	Treat no-gate as declared reference policy; report quality and compute separately.
Delta_t membership shortcut	Group membership alone fixes the sign for three of four t-strata and the target is structurally zero for unselected outsiders.	Report four-stratum metrics, a stratum-mean baseline and macro-averaged errors; pooled metrics secondary; validate signs conditionally on world_id.
Dataset/model incompatibility	Robust04 long documents and no default dense index can multiply scoring cost or block the semantic retrieval comparison.	Use TREC DL/MS MARCO passages as primary; require an artifact/length-feasibility gate; limit Robust04 to a small secondary study with fixed segmentation/aggregation.
Hashing scope/cost	A hash representation may require an expensive encoder pass and expand scope without adding useful prediction.	Conventional baseline first; hashing optional; keep only if the exact method and a cost-fidelity gain are demonstrated.

15.1 What would constitute success?

A strong result would show that (i) the intervention engine produces correct effects on synthetic cases; (ii) varying t reveals meaningful budget/utility trade-offs on real data; (iii) conventional features predict effects beyond a strata-only baseline; (iv) query-level attribution identifies which queries benefit from outsider allocation; and (v) findings differ systematically across controlled stage pairings. Hash features add value only if they improve the measured cost-fidelity frontier. A null result is still useful if the benchmark is rigorous and shows where cheap prediction fails, but publication value depends on reproducibility, sound comparisons and a clear empirical finding.

15.2 Novelty positioning

Do not claim that counterfactual explanation, stage ablation, cascaded retrieval or hashing is new by itself. Retrieval-conditioned reranking [24] and causal mediation for neural NLP [25] are relevant adjacent precedents, but neither is the same as the proposed document-level pipeline intervention estimands. The intended contribution is the specific

combination of document-level gate/reranker effects in a fixed Top-K cascade, budget-controlled candidate replacement with t, and a benchmark of inexpensive logged-trace features for predicting effects across stage-mismatch conditions. Hashing is an optional implementation, not a novelty claim.

16. Minimum Viable Project and Final Checklist

The minimum viable version is: TREC DL/MS MARCO passage benchmark subject to qrels/artifact feasibility checks; BM25 -> pointwise MiniLM as the core cascade; one dense-retriever comparison; a fixed universe; validated actual/no-reranker worlds; a full no-gate reference for a feasible subset; t-grid experiments under B=Kc; conventional prediction baselines; query-level train/test split; qrels-coverage and cost reporting. Hashing and Robust04 are optional extensions, not prerequisites.

- Freeze dataset and query splits only after checking qrel coverage in U, C and D.
- Freeze U construction, M and the contribution of every core retriever to the universe.
- Freeze Kc, B, t grid and outsider-selection policies; ensure t=1 is the candidate-only baseline when B=Kc.
- Implement and test actual, no-reranker, no-gate and t worlds separately.
- Freeze utility, cutoff, qrels policy, unjudged-document treatment and tie breaking.
- Freeze reranker checkpoint and prove pointwise score invariance to input order.
- If hashing is retained, confirm the exact method (LSH/LEH/other), input representation, bit length and feature-generation cost before adding it to the core experimental table.
- Pre-register primary and optional mismatch axes; only alter one major axis at a time.
- Report E_gate separately for survivors/non-survivors and Delta_t separately across its four membership-defined strata; include a strata-only baseline and do not use pooled correlation as the primary t-effect result.
- Provide end-to-end run manifests, seeds, software versions, cached scores and reproducibility instructions.

Final summary Measure the effects of gating and reranking; use t to probe candidate allocation at fixed reranker budget; use conventional and LSH/neural-hashing features to predict reference effects; and test whether these patterns change across lexical, semantic and cost-related stage pairings.

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